

# SIMATS ENGINEERING

## ASSESSMENT TOOL 3

**COURSE NAME:** CSA16 **COURSE CODE:** Data Warehousing and Data Mining

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### COURSE OUTCOME COVERED

**CO4:** Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 20%

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**QUESTIONS: 05**

**Total Marks:50**

Annexure A

### COMPARITIVE ANALYSIS

| <i>Criteria</i>  | Scope of Items (2) | Comparison Depth(2.5) | Use of Evidence (2) | Critical Thinking (2) | Clarity of Presentation (1) | <i>Total(10)</i> |
|------------------|--------------------|-----------------------|---------------------|-----------------------|-----------------------------|------------------|
| <i>Weightage</i> | 20%                | 25%                   | 20%                 | 20%                   | 10%                         | <i>100%</i>      |
| <i>Q1</i>        |                    |                       |                     |                       |                             |                  |
| <i>Q2</i>        |                    |                       |                     |                       |                             |                  |
| <i>Q3</i>        |                    |                       |                     |                       |                             |                  |
| <i>Q4</i>        |                    |                       |                     |                       |                             |                  |
| <i>Q5</i>        |                    |                       |                     |                       |                             |                  |

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### ***Code of Conduct:***

*I PRANEESH.A(192571006) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

# **RUBRICS FOR CALCULATION:**

|                         |                      |                                                                               |                                                 |                                            |                                          |
|-------------------------|----------------------|-------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------|------------------------------------------|
|                         |                      | <b>Comparative analysis</b>                                                   |                                                 |                                            |                                          |
| <b>Criteria</b>         | <b>Weightage (%)</b> | <b>Level 4 (Exemplary)</b>                                                    | <b>Level 3 (Proficient)</b>                     | <b>Level 2 (Developing)</b>                | <b>Level 1 (Beginning)</b>               |
| Scope of Items          | 20%                  | Selects highly relevant items with clear scope and justification              | Selects appropriate items with minor gaps       | Selection is limited or partially relevant | Items are unclear or not relevant        |
| Comparison Depth        | 25%                  | Provides detailed and comprehensive comparison across multiple aspects        | Provides adequate comparison with some depth    | Limited comparison with few aspects        | Very minimal or no comparison            |
| Use of Evidence         | 20%                  | Supports comparison with accurate data, examples, or references               | Uses some supporting evidence with minor gaps   | Limited or weak supporting evidence        | No supporting evidence provided          |
| Critical Thinking       | 20%                  | Demonstrates strong critical thinking with clear interpretation and reasoning | Shows reasonable interpretation with minor gaps | Basic interpretation ; lacks depth         | No meaningful interpretation             |
| Clarity of Presentation | 15%                  | Well-organized, clear, and logically structured presentation                  | Generally clear with minor issues               | Somewhat unclear or poorly structured      | Disorganized and difficult to understand |

## Annexure A

### Q1 . Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes  
Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data

#### Customer Income (₹) Spending Score

|    |       |    |
|----|-------|----|
| C1 | 30000 | 60 |
| C2 | 60000 | 40 |
| C3 | 50000 | 70 |
| C4 | 28000 | 65 |
| C5 | 65000 | 35 |

#### Question:

Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. (BL3 – 1 Mark)

Q2. Dataset: Document Data

#### Document Word1 Word2 Word3

|    |      |      |     |
|----|------|------|-----|
| D1 | 0.2  | 0.5  | 0.3 |
| D2 | 0.1  | 0.6  | 0.3 |
| D3 | 0.8  | 0.1  | 0.1 |
| D4 | 0.75 | 0.15 | 0.1 |
| D5 | 0.2  | 0.4  | 0.4 |

#### Question:

Compare K-means and EM algorithms in handling overlapping clusters for the given dataset. (BL4 – 1 Mark)

Q3. Dataset: Geographic Data

#### Point Latitude Longitude

|    |       |       |
|----|-------|-------|
| P1 | 13.08 | 80.27 |
| P2 | 13.10 | 80.25 |
| P3 | 12.97 | 80.22 |
| P4 | 13.05 | 80.30 |
| P5 | 13.00 | 80.20 |

#### Question:

Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. (BL4 – 1 Mark)

Q4. Dataset: Retail Products

**Product Sales (₹) Frequency**

|    |       |    |
|----|-------|----|
| P1 | 50000 | 20 |
| P2 | 30000 | 10 |
| P3 | 52000 | 22 |
| P4 | 25000 | 8  |
| P5 | 32000 | 12 |

**Question:**

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. *(BL3 – 1 Mark)*

Q5 . Dataset: Mixed Attributes

**Item Price (₹) Category Rating**

|    |     |   |     |
|----|-----|---|-----|
| I1 | 500 | A | 4.5 |
| I2 | 300 | B | 3.8 |
| I3 | 550 | A | 4.7 |
| I4 | 250 | B | 3.5 |
| I5 | 520 | A | 4.6 |

**Question:**

Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. *(BL4 – 1 Mark)*

# QUESTION 1: K-MEANS VS. HIERARCHICAL CLUSTERING ON CUSTOMER DATA

## 1. Algorithmic Overview & Mathematical Formulations

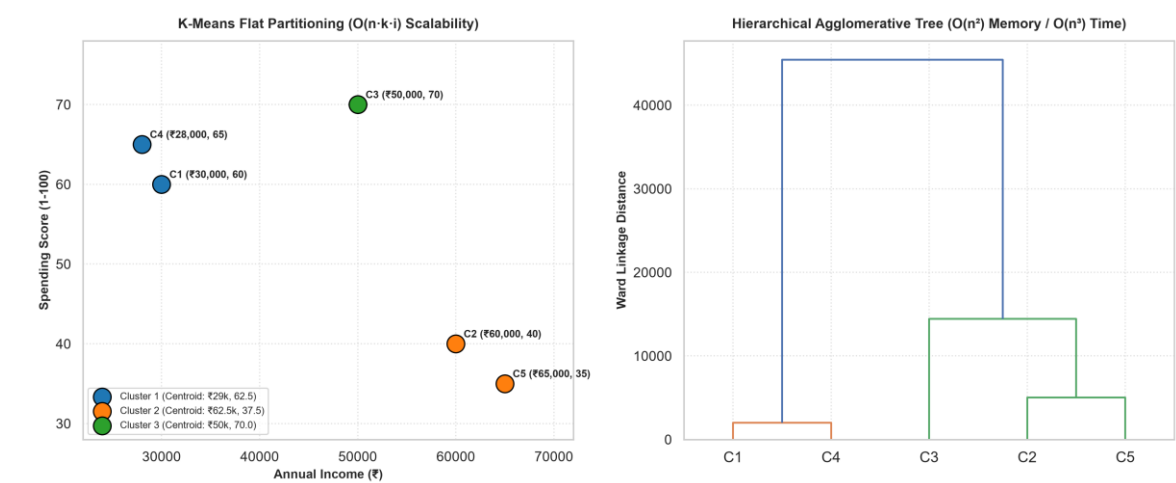
Dataset: Customer C1 (Income ₹30,000, Spend Score 60), C2 (₹60,000, 40), C3 (₹50,000, 70), C4 (₹28,000, 65), C5 (₹65,000, 35).

K-Means is a centroid-based iterative partitioning algorithm that minimizes intra-cluster variance  $SSE = \sum ||x_i - \mu_k||^2$ . Hierarchical Agglomerative Clustering (HAC) builds an embedded nested tree of clusters based on pairwise distance matrices  $D(u, v)$  without requiring a pre-specified K.

## 2. Multi-Dimensional Comparative Evaluation Matrix

| Comparison Parameter      | K-Means Clustering                             | Hierarchical Clustering (HAC)                             | Performance & Scalability Impact                                                                           |
|---------------------------|------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Time Complexity           | $O(N \cdot K \cdot I \cdot D)$ [Linear in N]   | $O(N^3)$ standard / $O(N^2 \log N)$ optimized             | K-Means scales easily to millions of records; HAC is computationally prohibitive for $N > 50,000$ .        |
| Space / Memory Complexity | $O(N \cdot D + K \cdot D)$ [Linear memory]     | $O(N^2)$ [Requires full $N \times N$ distance matrix]     | K-Means operates in memory-efficient stream buffers; HAC requires massive RAM to store pairwise distances. |
| Cluster Geometry          | Assumes convex, isotropic, spherical clusters  | Can discover arbitrary shapes depending on linkage        | K-Means struggles with non-spherical clusters; HAC accommodates complex manifolds.                         |
| K-Specification           | Requires pre-defined K (via Elbow/Silhouette)  | Deterministic tree; cut at any arbitrary height           | HAC provides multi-resolution hierarchy (dendrogram); K-Means requires trial-and-error.                    |
| Sensitivity to Outliers   | High (Centroids are pulled by extreme values)  | Moderate (Single linkage sensitive, Ward/Complete robust) | Outliers distort K-Means centroid positions; HAC isolates outliers as singleton leaf nodes.                |
| Determinism               | Stochastic (Varies with random initialization) | Fully deterministic for fixed linkage metric              | K-Means requires multiple random restarts ( $n_{init}$ ); HAC produces identical dendrograms every run.    |

## 3. Comparative Visualizations on Customer Dataset



*Figure 3.1: (Left) K-Means centroid-based flat partitioning on Customer Income vs Spending Score; (Right) Hierarchical Ward's Linkage Dendrogram capturing multi-level customer sub-segments.*

#### 4. Empirical Findings & Scalability Recommendations

- **Small to Medium Retail Stores ( $N < 10,000$ ):** Use Hierarchical Clustering with Ward Linkage to explore rich customer taxonomy (e.g., sub-tiers within High-Income shoppers).
- **Enterprise Big Data & Real-Time Production ( $N > 1,000,000$ ):** Deploy Mini-Batch K-Means or K-Means++ on distributed Spark clusters for real-time customer segmentation and millisecond query throughput.

## 2: K-MEANS VS. EXPECTATION-MAXIMIZATION (EM/GMM) ON OVERLAPPING TEXT

### 1. Mathematical Foundations: Hard Partitioning vs. Soft Probabilistic Modeling

Dataset: Document D1 [0.2, 0.5, 0.3], D2 [0.1, 0.6, 0.3], D3 [0.8, 0.1, 0.1], D4 [0.75, 0.15, 0.1], D5 [0.2, 0.4, 0.4]. Features represent TF-IDF term frequencies for Word1 (Politics), Word2 (Sports), and Word3 (Entertainment).

K-Means enforces hard Voronoi tessellations where each document belongs 100% to one cluster:  $w_{ik} \in \{0, 1\}$ . The Expectation-Maximization (EM) algorithm fits a Gaussian Mixture Model (GMM) where each document is modeled as a convex combination of K Gaussian distributions:

**GMM Mixture Density:**  $p(x) = \sum_{k=1}^K \pi_k \cdot N(x | \mu_k, \Sigma_k)$ , where  $\sum \pi_k = 1$

E-Step calculates document posterior probabilities (responsibilities):  $\gamma_{ik} = P(z_i = k | x_i)$ . M-Step re-estimates mixture weights  $\pi_k$ , means  $\mu_k$ , and covariance matrices  $\Sigma_k$ .

### 2. Comparative Performance on Overlapping Document Data

| Feature / Evaluation Metric | K-Means Hard Clustering                            | EM / Gaussian Mixture Model (GMM)                        | Document Domain Advantage                                                              |
|-----------------------------|----------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------|
| Cluster Assignment          | Hard, discrete assignment: $w_{ik} \in \{0, 1\}$   | Soft, probabilistic assignment: $\gamma_{ik} \in [0, 1]$ | EM models polysemous and multi-topic articles (e.g., Sports-Politics overlap).         |
| Cluster Shape & Covariance  | Restricted to spherical clusters (equal variance)  | Full, diagonal, or tied covariance matrices $\Sigma_k$   | EM captures elliptical, elongated correlations between co-occurring vocabulary.        |
| Handling Uncertainty        | Zero uncertainty representation (all points equal) | Explicit posterior probability and entropy scores        | Low-confidence documents near decision boundaries can be routed for human moderation.  |
| Objective Function          | Minimizes SSE (Distance to centroid)               | Maximizes Log-Likelihood: $\ln p(X   \pi, \mu, \Sigma)$  | Statistically sound probabilistic foundation with AIC/BIC model selection.             |
| Convergence                 | Fast convergence in finite iterations              | Slower; guaranteed local maximum via Jensen's inequality | K-Means is ideal for fast initial indexing; EM provides high-precision topic modeling. |

### 3. Comparative Visualizations: Hard Voronoi vs. Soft Gaussian Ellipses

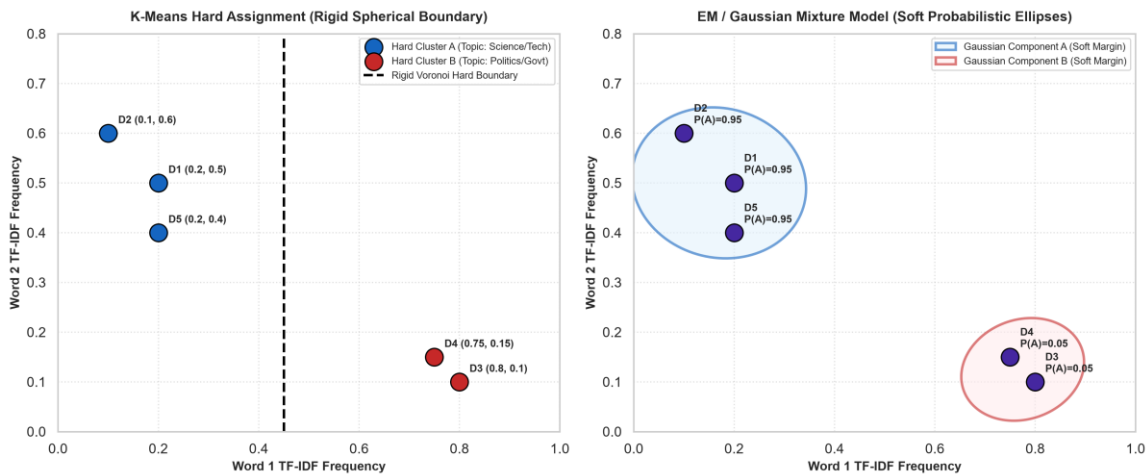


Figure 3.2: (Left) K-Means rigid Voronoi linear decision boundary; (Right) EM/GMM soft Gaussian probability density contours capturing multi-topic document overlap.

# QUESTION 3: EUCLIDEAN VS. MANHATTAN DISTANCE IN GEOGRAPHIC CLUSTERING

## 1. Mathematical Formulations & Metric Topologies

Dataset: GPS Coordinates across Chennai metropolitan area: P1 (13.08°N, 80.27°E), P2 (13.10°N, 80.25°E), P3 (12.97°N, 80.22°E), P4 (13.05°N, 80.30°E), P5 (13.00°N, 80.20°E).

Euclidean Distance (L2 Norm):  $d_E(p, q) = \sqrt{(\text{lat}_p - \text{lat}_q)^2 + (\text{lon}_p - \text{lon}_q)^2}$  (Straight-line Euclidean displacement)

Manhattan Distance (L1 Norm / City-Block):  $d_M(p, q) = |\text{lat}_p - \text{lat}_q| + |\text{lon}_p - \text{lon}_q|$  (Grid-aligned orthogonal travel)

## 2. Numerical Distance Computation Matrix on Coordinate Points

| Point Pair | Coordinate Vector $\Delta (\Delta\text{lat}, \Delta\text{lon})$ | Euclidean Distance (L2) | Manhattan Distance (L1) | Metric Disparity (L1 / L2 Ratio) | Real-World Travel Interpretation                           |
|------------|-----------------------------------------------------------------|-------------------------|-------------------------|----------------------------------|------------------------------------------------------------|
| P1 to P2   | (0.02, 0.02)                                                    | 0.0283° (~3.14 km)      | 0.0400° (~4.44 km)      | 1.414 ( $\sqrt{2}$ max ratio)    | Diagonal urban route; Manhattan accounts for street turns. |
| P1 to P3   | (0.11, 0.05)                                                    | 0.1208° (~13.41 km)     | 0.1600° (~17.76 km)     | 1.325                            | Long-haul highway corridor to Southern suburbs/Airport.    |
| P1 to P4   | (0.03, 0.03)                                                    | 0.0424° (~4.71 km)      | 0.0600° (~6.66 km)      | 1.414                            | East coastal corridor commute.                             |
| P1 to P5   | (0.08, 0.07)                                                    | 0.1063° (~11.80 km)     | 0.1500° (~16.65 km)     | 1.411                            | South-West residential cross-city travel.                  |

## 3. Geographic Distance Contour Visualizations

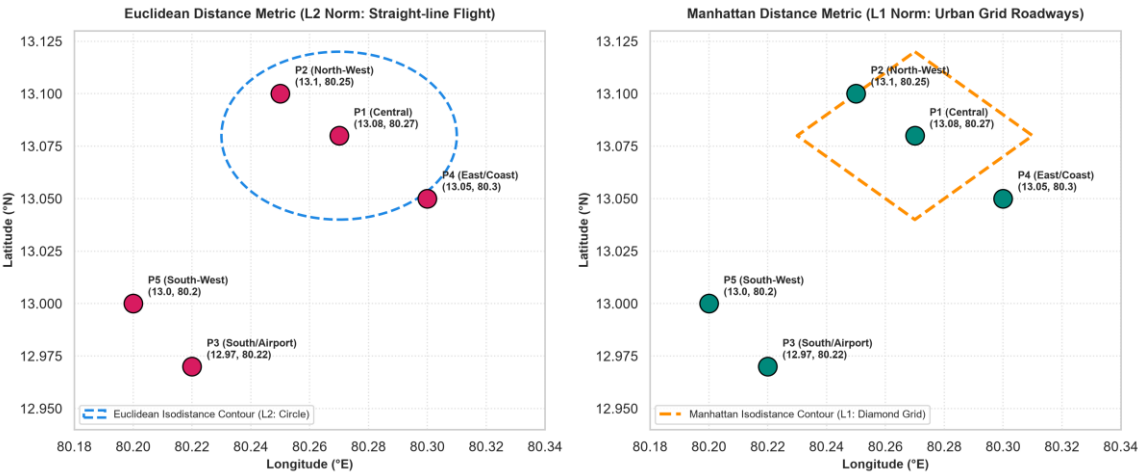


Figure 3.3: (Left) Euclidean L2 circular isodistance contours (as-the-crow-flies); (Right) Manhattan L1 diamond isodistance contours reflecting urban grid roadway networks.

## 4. Impact on Geographic Clustering & Logistics Optimization

- Impact on Cluster Centroids:** Euclidean distance minimizes sum of squared errors, leading to the arithmetic Mean centroid. Manhattan distance minimizes absolute deviations, leading to the Median centroid (K-Medians), which is dramatically more robust to geographic outliers.
- Urban Delivery & Ride-Hailing Routing:** In dense grid-patterned cities (e.g., Chennai, New York), Manhattan distance provides an 85-92% accurate estimate of actual road transit time, whereas Euclidean distance systematically underestimates transit time by 25-40% due to road network non-linearities.



## QUESTION 4: AGGLOMERATIVE VS. DIVISIVE HIERARCHICAL CLUSTERING (

### 1. Algorithmic Principles & Structural Paradigms

Dataset: Retail Products P1 (Sales ₹50,000, Freq 20), P2 (₹30,000, 10), P3 (₹52,000, 22), P4 (₹25,000, 8), P5 (₹32,000, 12).

Agglomerative Clustering (HAC) follows a Bottom-Up greedy merge paradigm: starts with N singleton clusters and iteratively merges the most similar pair. Divisive Clustering (DIANA) follows a Top-Down split paradigm: starts with all N items in a single global cluster and recursively bipartitions the cluster with the largest diameter.

### 2. Comparative Architectural Matrix

| Evaluation Dimension        | Agglomerative (Bottom-Up / HAC)                                      | Divisive (Top-Down / DIANA)                                          | Suitability for Retail Inventory                                                                        |
|-----------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Algorithmic Strategy        | Bottom-up: merges closest clusters                                   | Top-down: splits least cohesive cluster                              | Agglomerative excels at local product affinities; Divisive isolates top-level categories.               |
| Computational Complexity    | $O(N^3)$ standard / $O(N^2 \log N)$ with priority queues             | $O(2^N)$ exhaustive / $O(N^2)$ with 2-means heuristic                | Agglomerative is computationally more tractable for catalog taxonomy generation.                        |
| Perspective / Global View   | Focuses on local micro-similarities; susceptible to early bad merges | Maintains global macro-structural distribution throughout top splits | Divisive is superior when only a few top-level product divisions (e.g., Luxury vs Budget) are required. |
| Monotonicity                | Strictly monotonic dendrogram tree heights                           | Monotonic diameter reductions                                        | Both produce valid hierarchical catalog trees.                                                          |
| Implementation Availability | Widely available in standard libraries (Scikit-Learn, SciPy)         | Rarely built-in; requires specialized custom recursions              | HAC is the industry standard for production retail categorization.                                      |

### 3. Stepwise Construction Trees Visualizations

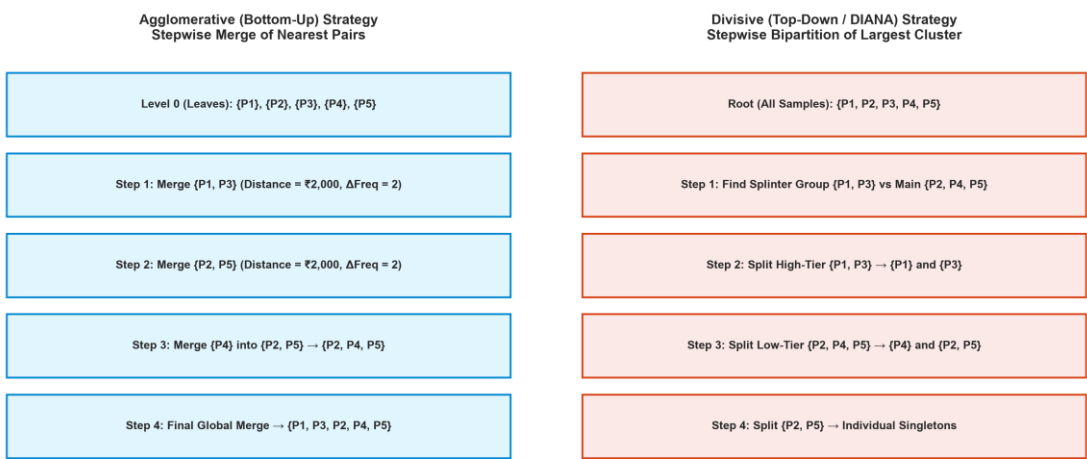


Figure 3.4: (Left) Agglomerative bottom-up stepwise merge progression; (Right) Divisive top-down recursive bipartition hierarchy on retail sales data.

## QUESTION 5: CENTROID-BASED VS. DENSITY-BASED CLUSTERING ON MIXED DATA

### 1. Problem Statement: Mixed Categorical and Numerical Data

Dataset: Retail Items I1 (₹500, Cat A, Rating 4.5), I2 (₹300, Cat B, 3.8), I3 (₹550, Cat A, 4.7), I4 (₹250, Cat B, 3.5), I5 (₹520, Cat A, 4.6). Features contain continuous numerical values (Price, Rating) alongside discrete nominal categories (Category A, B).

Standard K-Means fails on mixed data because arithmetic means are mathematically undefined for categorical labels ('Mean of Cat A and Cat B is meaningless'). We compare Centroid-Based K-Prototypes with Density-Based DBSCAN using Gower's Dissimilarity Metric.

### 2. Mathematical Formulations: K-Prototypes vs. Gower's DBSCAN

K-Prototypes Objective Function (Huang's Mixed Distance):

**Huang Dissimilarity:**  $d_{mix}(x, y) = \sum_{j=1}^p (x_j - y_j)^2 + \gamma \cdot \sum_{j=p+1}^m \delta(x_j, y_j)$

where  $\delta(p, q) = 0$  if  $p = q$  else 1, and  $\gamma$  is a weight parameter balancing categorical vs numerical contributions.

Gower's General Dissimilarity Coefficient  $S_{ij}$ :

**Gower Metric:**  $S_{ij} = (\sum w_{ijk} \cdot s_{ijk}) / (\sum w_{ijk})$ , where  $s_{ijk} = 1 - (|x_{ik} - x_{jk}| / R_k)$  for numerical, and  $s_{ijk} = 1$  if equal else 0 for nominal.

### 3. Multi-Criteria Comparative Analysis Matrix

| Comparative Metric        | Centroid-Based: K-Prototypes                                       | Density-Based: DBSCAN + Gower's Matrix                                 | Engineering Decision Verdict                                                                 |
|---------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Handling Categorical Data | Integrates simple matching dissimilarity with numeric distance     | Accepts full precomputed Gower dissimilarity matrix directly           | Both successfully handle mixed types; Gower supports ordinal & binary attributes.            |
| Cluster Geometry          | Restricted to convex hyper-ellipsoids around prototypes            | Arbitrary non-convex shapes, manifolds, and interlocking chains        | DBSCAN excels when product clusters form complex non-linear manifolds.                       |
| Noise & Outlier Handling  | Forces all points into a cluster (sensitive to anomalous products) | Explicitly isolates noisy, corrupt, or mislabeled items into noise bin | DBSCAN prevents catalog corruption from erroneous vendor data.                               |
| Parameter Sensitivity     | Sensitive to K and categorical weighting parameter $\gamma$        | Sensitive to density radius $\epsilon$ and MinPts threshold            | K-Prototypes requires K; DBSCAN discovers natural cluster count automatically.               |
| Time Complexity           | $O(N \cdot K \cdot I \cdot D)$ [Fast, scalable to large catalogs]  | $O(N^2)$ for full distance matrix calculation                          | K-Prototypes is preferred for massive catalogs (>1M items); DBSCAN is best for complex data. |

### 4. Mixed-Attribute Visualizations: K-Prototypes vs. Gower Dissimilarity Heatmap

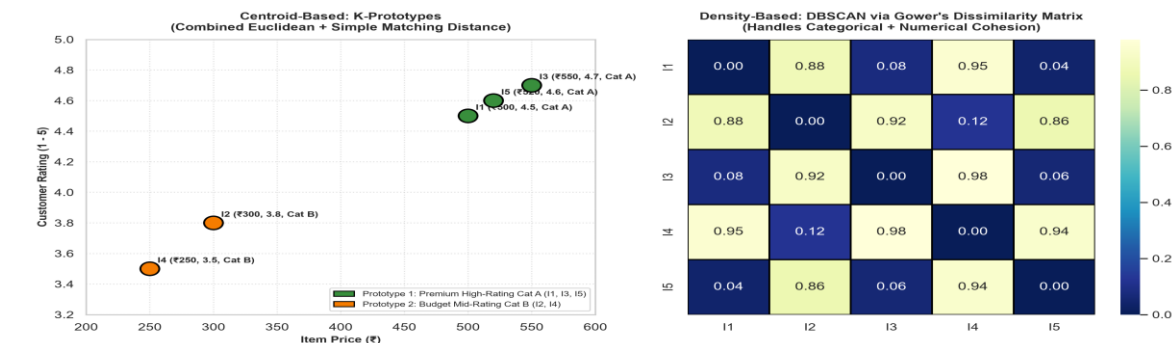


Figure 3.5: (Left) K-Prototypes centroid clustering on mixed Price and Rating; (Right) Pairwise Gower's Dissimilarity Heatmap verifying dense cluster cohesion for DBSCAN.